

The Operator Algebra of the Emission Semiring

A λ -Ring, the Adams Square, and Two Characters

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A Companion to *Lehmer's Box*, *The Emission–Gap Theorem*,
The Occupant of the Salem Slot, and *The Generative Content of a Conserved Emptiness*

Abstract

The companion papers treat the spectral operators $\oplus$ (direct sum), $\otimes$ (tensor), and $(\cdot)^2$ (squaring) one at a time. This note treats them *in relation* — to each other and to themselves — and finds that they are not an ad hoc collection but a λ -ring. The pair $(\oplus, \otimes)$ is a commutative semiring with $\otimes$ distributing over $\oplus$; the squaring operator is the Adams operation ψ^2 , a semiring endomorphism tied to the exterior powers by Newton's identity, and equal to the diagonal of the tensor square. Two characters resolve the structure completely. The Mahler measure is *additive* on $\oplus$ — a monoid homomorphism onto $\langle \varphi, 2, 3, 5, \beta^2 \rangle$ whose least generator is the cost floor φ — but *tropical* on $\otimes$: $\log M(A \otimes B)$ is a $(\max, +)$ convolution of the log-spectra, never a product. The angle charge is a finite character onto $\mathbb{Z}/4\mathbb{Z}$, the sumset under $\otimes$ and doubling under ψ^2 . The three self-iterations are the structure's three powers: additive ($M = \varphi^k$, linear), Adams ($M = \varphi^{2^k}$, a doubling tower at fixed degree), and multiplicative ($M = \varphi^{S_k}$ with $S_k = k(\lfloor (k-1)/2 \rfloor)$, super-exponential). Every relation below is machine-verified in exact arithmetic; the backing check identifiers are named inline.

1 Introduction: the operators in relation

The construction analysed in the companion papers grows an exact symbolic state by applying a fixed set of operators to a seed catalogue. The *spectral* operators — those whose effect on the output spectrum is forced by the inputs — are direct sum $\oplus$, tensor (Kronecker) product $\otimes$, squaring $(\cdot)^2$, and the spectrum-preserving minpoly and Φ . Each was studied in isolation: $\oplus$ multiplies Mahler measures, $\otimes$ adds angle charge, $(\cdot)^2$ doubles it. The question this note answers is structural: *how do these operators relate to one another, and to themselves?*

The answer is a single named object. We show (§2–3) that $(\oplus, \otimes)$ make the algebra's spectra a commutative semiring and that squaring is the Adams operation ψ^2 of the associated λ -ring — a semiring endomorphism, the Newton power map, and the diagonal of the tensor square, all at once. We then show (§4–5) that the two invariants the companion papers use are exactly the two characters of this structure: the Mahler measure, additive on $\oplus$ and *tropical* on $\otimes$, and the angle charge, a finite character onto $\mathbb{Z}/4\mathbb{Z}$. The three ways an operator can act on itself (§6) are the three powers of the semiring, with the rates already recorded. Fixed points and the spectrum-preserving operators close the picture (§7). The unifying statement (§8) is that the floor φ and the superselection charge are not two facts but one λ -ring seen through its two characters.

2 The emission semiring

Definition 2.1 (Objects and operators). An *object* is a finite multiset $A = \{\lambda_1, \dots, \lambda_d\}$ of nonzero algebraic numbers — the eigenvalue multiset of a catalogue matrix or any object built from them. Write $\mathcal{S}$ for the set of objects. Define

$$A \oplus B = A \uplus B \quad (\text{multiset union}), \quad A \otimes B = \{\lambda\mu : \lambda \in A, \mu \in B\} \quad (\text{with multiplicity}),$$

realised on matrices by block diagonal sum and Kronecker product. Let $\mathbf{0} = \emptyset$ and $\mathbf{1} = \{1\}$.

These are the spectra of $\oplus$ and $\otimes$: $\text{spec}(A \oplus B) = \text{spec } A \uplus \text{spec } B$ and $\text{spec}(A \otimes B) = \{\lambda\mu\}$. The structure is the positive cone of the monoid ring $\mathbb{N}[\mathbb{C}^\times]$ restricted to the eigenvalue group the catalogue generates.

Theorem 2.2 (The emission semiring). $(\mathcal{S}, \oplus, \otimes, \mathbf{0}, \mathbf{1})$ is a commutative semiring: $\oplus$ and $\otimes$ are commutative and associative, $\otimes$ distributes over $\oplus$,

$$A \otimes (B \oplus C) = (A \otimes B) \oplus (A \otimes C),$$

$\mathbf{0}$ is the additive identity and a multiplicative annihilator, and $\mathbf{1}$ is the multiplicative identity. *[forced]* (OA-SR-01..08)

Proof. Multiset union is commutative, associative, and has identity $\emptyset$. The elementwise product is commutative and associative because multiplication in $\mathbb{C}^\times$ is. Distributivity is the multiset identity $\{\lambda\nu : \lambda \in A, \nu \in B \uplus C\} = \{\lambda\mu\} \uplus \{\lambda\rho\}$ over $\mu \in B, \rho \in C$. The identity $\{1\}$ fixes every object under $\otimes$, and $\emptyset \otimes A = \emptyset$. $\square$

The Mahler measure and the angle charge will be the two morphisms *out* of this semiring. Before defining them we identify the third operator.

3 The Adams square

Definition 3.1 (Power operations). For $n \geq 1$ let $\psi^n(A) = \{\lambda^n : \lambda \in A\}$ with multiplicity. The squaring operator of the companion papers is $(\cdot)^2 = \psi^2$.

Theorem 3.2 (Squaring is the Adams operation). Each ψ^n is a semiring endomorphism of $(\mathcal{S}, \oplus, \otimes)$:

$$\psi^n(A \oplus B) = \psi^n A \oplus \psi^n B, \quad \psi^n(A \otimes B) = \psi^n A \otimes \psi^n B, \quad \psi^n(\mathbf{1}) = \mathbf{1},$$

and they compose by $\psi^m \circ \psi^n = \psi^{mn}$. On power sums $p_n = \text{tr } \psi^n$ they obey Newton's identities; in particular

$$\text{tr } \psi^2 A = p_2 = e_1^2 - 2e_2,$$

the relation that expresses ψ^2 through the exterior powers $e_1 = \text{tr } A$, $e_2 = \text{tr } \Lambda^2 A$. *[forced]* (OA-PSI-01..05)

Proof. ψ^n acts on eigenvalues by $\lambda \mapsto \lambda^n$. Under $\oplus$ this permutes the union; under $\otimes$, $(\lambda\mu)^n = \lambda^n \mu^n$, so $\psi^n(A \otimes B) = \psi^n A \otimes \psi^n B$. Composition is $(\lambda^n)^m = \lambda^{nm}$. Newton's identity $p_2 = e_1^2 - 2e_2$ is the case $k = 2$ of $p_k = e_1 p_{k-1} - \dots \pm k e_k$, verified symbolically for the catalogue degrees. $\square$

That ψ^2 is multiplicative as well as additive is what makes the operator set a λ -ring rather than merely a semiring with an extra unary map: in the Grothendieck completion $K(\mathcal{S}) = \mathbb{Z}[\text{eigenvalue group}]$ the Adams operations ψ^n are precisely the ring endomorphisms determined by the λ -operations (exterior powers) through Newton's identities (Grothendieck; Adams; Atiyah; Knutson). The next proposition gives the geometric content of ψ^2 .

Proposition 3.3 (Plethysm: the square is a diagonal). *For every object A ,*

$$A \otimes A = \text{Sym}^2 A \oplus \Lambda^2 A, \quad \text{Sym}^2 A = \{\lambda_i \lambda_j : i \leq j\}, \quad \Lambda^2 A = \{\lambda_i \lambda_j : i < j\},$$

and $\psi^2 A = \{\lambda_i^2\}$ is exactly the diagonal $i = j$ inside $\text{Sym}^2 A$. Their traces are

$$\text{tr Sym}^2 A = \frac{1}{2}(p_1^2 + p_2), \quad \text{tr } \Lambda^2 A = \frac{1}{2}(p_1^2 - p_2) = e_2.$$

[forced] (OA-PLE-01..06)

Thus the squaring operator is not a fourth primitive: it is the diagonal restriction of the tensor square, the off-diagonal part being the exterior square Λ^2 . Iterating, $\psi^{2^k} = (\psi^2)^{\circ k}$ sends each λ to λ^{2^k} (OA-PSI-05).

4 Character I: the Mahler measure

Definition 4.1. $M(A) = \prod_{\lambda \in A} \max(1, |\lambda|)$, the Mahler measure of the object.

Theorem 4.2 (The measure is additive on $\oplus$ and tropical on $\otimes$). *The Mahler measure satisfies*

$$M(A \oplus B) = M(A) M(B), \quad M(\psi^2 A) = M(A)^2,$$

so it is a monoid homomorphism $(\mathcal{S}, \oplus) \rightarrow (\mathbb{R}_{\geq 1}, \cdot)$ intertwining ψ^2 with squaring. On the tensor it is not multiplicative; instead, writing $x^+ = \max(0, x)$,

$$\log M(A \otimes B) = \sum_{\lambda \in A} \sum_{\mu \in B} (\log |\lambda| + \log |\mu|)^+,$$

a $(\max, +)$ (tropical) convolution of the log-spectra. [forced] (OA-M-01..07)

Proof. For $\oplus$ the root multiset is the union and $\max(1, \cdot)$ is multiplicative across it. For ψ^2 , $\max(1, |\lambda|^2) = \max(1, |\lambda|)^2$. For $\otimes$ the eigenvalues are the products $\lambda\mu$, so $\log M = \sum \log^+ |\lambda\mu| = \sum (\log |\lambda| + \log |\mu|)^+$. Non-multiplicativity is exhibited by golden $\otimes (x^2 - 2)$: $M = 2\varphi^2 = 5.2360\dots$ while $M(\varphi)M(x^2 - 2) = 2\varphi = 3.2360\dots$ $\square$

So the two ring operations map under $\log M$ to two different arithmetics: $\oplus$ to ordinary addition, $\otimes$ to the tropical semiring. The image of the additive part is the object of the companion note on the spectrum.

Theorem 4.3 (The $\oplus$ -image is the spectrum monoid; the floor is its least generator). $M(\mathcal{S}, \oplus) = \langle \varphi, 2, 3, 5, \beta^2 \rangle$ multiplicatively, with the single relation $\beta^2 = \varphi^2 \sqrt{5}$. The generators $\varphi, 2, 3, 5$ are multiplicatively independent (norms $-1, 4, 9, 25$ in $\mathbb{Q}(\sqrt{5})$); the monoid has five atoms but rank four, hence is not factorial, with $\beta^2 \cdot \beta^2 = 5 \cdot \varphi^4$. Its least generator is φ , so the monoid contains no element in $(1, \varphi)$ — the cost floor. *[forced/computed] (OA-M-08; GA-01..13)*

The cost floor of the growth rule is therefore an intrinsic feature of Character I: it is the smallest generator of the additive character's image, and $\lambda \log \varphi$ is λ times that generator.

5 Character II: the angle charge

Definition 5.1. For an object whose eigenvalues lie at arguments in $\frac{\pi}{2}\mathbb{Z}$, let $\chi(A)$ be the multi-set of charges $q = \lfloor 2 \arg \lambda / \pi \rfloor \bmod 4 \in \mathbb{Z}/4\mathbb{Z}$.

Theorem 5.2 (The charge is a finite character). *On the emission algebra the angle charge transforms by*

$$\chi(A \otimes B) = \chi(A) + \chi(B) \text{ (sumset mod 4)}, \quad \chi(\psi^2 A) = 2\chi(A), \quad \chi(A \oplus B) = \chi(A) \uplus \chi(B).$$

The full group $\mathbb{Z}/4\mathbb{Z}$ is realised: the real seeds carry $\{0, 2\}$, and the Lorentzian seed $K = x^4 + 5x^2 - 5$ carries $\{0, 1, 2, 3\}$, its imaginary pair $\pm i\beta$ at $\pm \frac{\pi}{2}$. [forced/computed] (OA-CH-01..05)

Proof. $\arg(\lambda\mu) = \arg \lambda + \arg \mu$ gives the sumset on $\otimes$; $\arg(\lambda^2) = 2 \arg \lambda$ gives doubling on ψ^2 ; $\oplus$ unions the spectra. The catalogue charges are computed directly. $\square$

Character II is the superselection rule of the companion paper: the Salem regime is the off-charge sector ($\arg \notin \frac{\pi}{2}\mathbb{Z}$), and because χ is preserved by $\oplus, \otimes, \psi^2$, no finite word reaches it. Where Character I records growth, Character II records the on-circle alphabet — and the doubling $\chi(\psi^2 A) = 2\chi A$ is the charge-level shadow of $M(\psi^2 A) = M(A)^2$.

6 Self-action: the three powers

An operator applied to itself produces one of three rates — the additive, Adams, and multiplicative powers of the semiring. From the golden seed (degree 2, $M = \varphi$):

power	degree(k)	$M(k)$	character of the rate
$\oplus^k$ (additive)	$2k$	φ^k	linear; $\log M / \deg = \frac{1}{2} \log \varphi$ constant
$(\psi^2)^{\circ k}$ (Adams)	2	φ^{2^k}	doubling tower at fixed degree
$\otimes^k$ (multiplicative)	2^k	φ^{S_k}	super-exponential

Proposition 6.1 (Rates). *With $S_k = k(\frac{k-1}{\lfloor (k-1)/2 \rfloor})$ ($= 1, 2, 6, 12, 30, 60, 140, \dots$), the k -fold direct sum, Adams iterate, and tensor power of the golden seed have the measures tabulated above; $S_k \sim 2^k \sqrt{k/2\pi}$, and S_k is the mean-absolute-deviation sum of the symmetric binomial. [forced/computed] (OA-IT-01..04; GR-table)*

The additive power holds a *constant* entropy density $\frac{1}{2} \log \varphi$ per dimension — the floor expressed as a rate — while the Adams and multiplicative powers raise it. The three are the same semiring's three notions of exponentiation: repeated addition, repeated Frobenius, repeated multiplication.

7 Fixed points and the spectrum-preserving operators

Proposition 7.1 (Adams-fixed measures are cyclotomic). *$M(\psi^2 A) = M(A)$ holds iff $M(A) = 1$, i.e. iff every eigenvalue of A is a root of unity or 0. The only measure fixed by squaring is the trivial one. [forced] (OA-FX-01)*

Proof. $M(A)^2 = M(A)$ forces $M(A) \in \{0, 1\}$; as $M \geq 1$, $M(A) = 1$, the Kronecker locus. $\square$

The operators minpoly and Φ (re-companion of the radical) act as the identity on the spectrum: they re-present an object without changing its eigenvalue multiset, hence fix both characters. They are the idempotents of the structure (OA-FX-02), and play no role in either the floor or the grading.

8 The unified picture

Theorem 8.1 (One λ -ring, two characters). *The spectral operators of the emission algebra form a commutative λ -ring: $(\oplus, \otimes)$ the semiring (Thm. 2.2), ψ^2 the Adams operation (Thm. 3.2, Prop. 3.3). The Mahler measure and the angle charge are its two characters: Character I (Thm. 4.2–4.3) is additive on $\oplus$ and tropical on $\otimes$, with image the spectrum monoid whose least generator is the floor φ ; Character II (Thm. 5.2) is finite, onto $\mathbb{Z}/4\mathbb{Z}$, the sumset on $\otimes$ and doubling on ψ^2 . The companion results are the two characters read off: the cost floor is Character I's least generator, the Salem-free emission is Character II's conserved sector, and $\sqrt{5} = \varphi + \varphi^{-1}$ is where the two meet — the $\otimes$ -logarithm of the floor and the grow generator of the self-action. [forced]*

What looked like three unrelated operators and two unrelated invariants is one algebraic object. The floor and the superselection charge are not independent theorems; they are the two ways a single λ -ring projects to a value monoid and to a finite group. The tropical face of the measure on $\otimes$ is the reason the floor survives composition: a $(\max, +)$ convolution of log-spectra bounded below by $\log \varphi$ stays bounded below by $\log \varphi$, no matter how deep the word.

9 Epistemic ledger

Statuses follow the companion discipline: [forced]/[computed] are proved here and re-derived in exact arithmetic; [declared] marks a framing. Backing identifiers index the machine checks (38 in the operator suite, plus the cited generator and rate suites; all green).

Claim	Status	Backing
$(\oplus, \otimes)$ a commutative semiring; $\otimes$ distributes over $\oplus$	[forced]	OA-SR-01..08
ψ^2 a semiring endomorphism; $\psi^m \circ \psi^n = \psi^{mn}$; Newton $p_2 = e_1^2 - 2e_2$	[forced]	OA-PSI-01..05
Plethysm $A \otimes A = \text{Sym}^2 A \oplus \Lambda^2 A$; ψ^2 the diagonal of Sym^2	[forced]	OA-PL-01..06
M additive on $\oplus$, $M(\psi^2 A) = M(A)^2$	[forced]	OA-M-01..02
$M(A \otimes B)$ tropical: $\log M = \sum(\log \lambda + \log \mu)^+$	[forced]	OA-M-03..07
$\oplus$ -image = $\langle \varphi, 2, 3, 5, \beta^2 \rangle$, $\beta^2 = \varphi^2 \sqrt{5}$; least generator φ	[forced/computed]	OA-M-08; GA-01..13
Charge a finite character: sumset on $\otimes$, doubling on ψ^2 , union on $\oplus$	[forced/computed]	OA-CH-01..05
Self-action rates: $\varphi^k, \varphi^{2^k}, \varphi^{S_k}, S_k = k(\lfloor \frac{k-1}{2} \rfloor)$	[forced/computed]	OA-IT-01..04
Adams-fixed measures = cyclotomic locus $M = 1$	[forced]	OA-FX-01
minpoly, Φ idempotent, spectrum- and character-preserving	[forced]	OA-FX-02
One λ -ring, two characters (synthesis)	[forced]	Thm. 8.1

Closing statement. The emission operators are a λ -ring, and the construction's two load-bearing invariants are its two characters. Reading the operators in relation rather than one at a time turns a list of separate facts — $\oplus$ multiplies measures, $\otimes$ adds charge, $(\cdot)^2$ doubles it, the floor is φ , no Salem is emitted — into a single statement with one structure behind it. The square is the Adams operation; the measure is additive then tropical; the charge is the finite character; and the floor is the least generator of the one and the fixed point of the other.

References

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All non-trivial claims were re-derived from definitions in exact arithmetic (`sympy/mpmath`; eigenvalue multisets to 45 digits). Citations are for orientation; bibliographic details should be confirmed before relying on them. This is a companion to the Echo–Squirrel emission–algebra stack.